

1. Sellmeier's Equation

a) Using the Taylor expansion, and considering only one resonance, we have that for $(\frac{\omega}{\omega_0})^2 \ll 1$,

$$n_{SE}^2(\omega) \approx 1 + B \left(1 + \frac{\omega^2}{\omega_0^2} \right)$$

b) After squaring and using the Taylor expansion again, we get (retaining the first and second term because the third term is negligibly small)

$$n_{toy}^2(\omega) \approx 1 + \frac{2c\gamma}{a\omega_0^2} \left(1 + \frac{\omega^2}{\omega_0^2} \right)$$

Which means that for the two models to be consistent,

$$B = \frac{2c\gamma}{a\omega_0^2}$$

c) We can divide the summation term by λ^2 to get,

$$n_{SE}^2(\lambda) = 1 + \sum_i \frac{B_i}{1 - \frac{C_i}{\lambda^2}}$$

This is exactly the same as Eq. (1) if we have the condition

$$\frac{C_i}{\lambda^2} = \frac{\omega^2}{\omega_i^2}$$

or,

$$C_i = \left(\frac{\omega\lambda}{\omega_i} \right)^2 = \left(\frac{2\pi c}{\omega_i} \right)^2$$

2. Discovering Evanescent Waves

a) From Snell's Law, we have

$$n_i \sin \theta_i = n_t \sin \theta_t$$

Rearranging gives us,

$$\sin \theta_t = \frac{n_i}{n_t} \sin \theta_i$$

and,

$$\cos \theta_t = \sqrt{1 - \left(\frac{n_i}{n_t} \sin \theta_i \right)^2}$$

b) Plugging this back into the expression $e^{i(\mathbf{k}_t \cdot \mathbf{r} - \omega t)}$ gives,

$$e^{i(k_{t,x}x + k_{t,y}y - \omega t)} = e^{i \cdot \pm i k_t \sqrt{(n_i \sin \theta_i / n_t)^2 - 1} y} e^{i(k_t \sin \theta_i x - \omega t)} = e^{-y/d} e^{i(k_t \sin \theta_i x - \omega t)}$$

with

$$\frac{1}{d} = k_t \sqrt{\left(\frac{n_i}{n_t} \sin \theta_i \right)^2 - 1}$$

$$d = \frac{1}{k_t \sqrt{\left(\frac{n_i}{n_t} \sin \theta_i \right)^2 - 1}} = \frac{c}{\omega n_t} \frac{1}{\sqrt{\left(\frac{n_i}{n_t} \sin \theta_i \right)^2 - 1}}$$

The other solution $e^{+y/d}$ is omitted because it doesn't make any physical sense, as the wave would be amplified.

c) If the low index layer thickness is small $l \ll d$, the evanescent field does not decay significantly before reaching the second interface. Leading to some of the light being transmitted instead of fully reflected